

ANTONIO ALFREDO MALAGUTTI

DATA, ARTIFICIAL INTELLIGENCE & MACHINE LEARNING SPECIALIST | PRINCIPAL DATA SCIENTIST /
SENIOR MLOPS ENGINEER

📍 São Paulo, SP, Brazil 📞 +55 (19) 99210-5415 ✉️ amalagutti@gmail.com 🔗 [linkedin.com/in/amalagutti](https://www.linkedin.com/in/amalagutti)

📄 EXECUTIVE SUMMARY

Data Engineering, Data Science, and Artificial Intelligence Specialist with a solid track record in developing and productionalizing (MLOps) complex predictive models, recommendation systems, and modern Generative AI architectures (LLMs, RAG, Fine-Tuning). Proficient in providing technical leadership for multidisciplinary data teams, establishing algorithmic governance, translating business requirements into data products, and optimizing elastic analytical infrastructure costs within cloud environments.

⚙️ KEY COMPETENCIES

- ✓ **Artificial Intelligence & Machine Learning:** Development of predictive models, classification, regression, and clustering algorithms, Deep Learning, and Natural Language Processing (NLP) utilizing Scikit-Learn, PyTorch, and TensorFlow.
- ✓ **Generative AI & Large Language Models (LLMs):** Implementation of RAG (Retrieval-Augmented Generation) architectures, advanced prompt engineering, fine-tuning open-source models, and workflow orchestration using LangChain and LlamaIndex alongside vector databases (Pinecone, Milvus, Chroma).
- ✓ **Data Engineering & Medallion Architecture:** Building scalable batch and streaming data pipelines, structuring Data Lakehouses (Databricks, Delta Lake), data modeling, and orchestrating analytical workflows with Apache Airflow, dbt, Apache Spark, and Kafka.
- ✓ **MLOps & Model Governance:** End-to-end model lifecycle management (MLOps governance), automated deployment via elastic APIs (FastAPI, Docker), experiment tracking and reproducibility with MLflow, and continuous monitoring of data drift and algorithmic bias.

👥 LEADERSHIP

- ✓ **Data Team Leadership & Mentorship:** Technical management of data scientists, data engineers, and data analysts, promoting agile methodologies customized for research and development (R&D) workflows.
- ✓ **Data Translation:** Acting as a bridge between executive decision-makers and highly technical engineering teams, converting business pain points into mathematical hypotheses and ROI-driven data products.
- ✓ **Data-Driven Culture Evangelization:** Disseminating practices of scientific experimentation, data governance, data quality assurance, and ethics regarding the application of artificial intelligence algorithms.
- ✓ **Complex AI Project Management:** End-to-end technical steering from conceptual proof of concept (PoC) to global-scale deployment, handling project risks and mitigating overengineering.
- ✓ **Cost Architecture & Governance (FinOps for AI):** Strategic planning of high-performance computing resources consumption (GPUs/TPUs) for model training and cloud-based inference.

📁 PROFESSIONAL EXPERIENCE

Principal Data Scientist / Generative AI Specialist

Leading Tech & IA Unicorn

2023 – Present

- ✓ Technical leadership in the architecture and implementation of an enterprise LLM intelligent assistant, leveraging advanced RAG to reduce complex customer support resolution times by 45%.
- ✓ Designed automated pipelines for Fine-Tuning and evaluation of open-source large language models (Llama, Mistral), optimizing token consumption costs and inference latency.
- ✓ Established the corporate vector database platform, enabling scalable semantic search across multimillion unstructured document repositories.
- ✓ Provided continuous technical mentorship to data science squads transitioning from traditional Machine Learning to advanced Deep Learning and Generative AI workflows.

Data Science & MLOps Tech Lead

2021 – 2023

Payments Fintech

- ✓ Conceived and implemented a unified MLOps platform leveraging MLflow and Kubernetes, reducing model deployment time-to-market from weeks to under 1 hour.
- ✓ Developed and refined fraud detection algorithms based on tabular neural networks and XGBoost, saving \$2.5M annually in operational losses while minimizing false-positive rates.
- ✓ Implemented strict Data Drift and Concept Drift monitoring routines, ensuring continuous model calibration and stability for credit underwriting models.
- ✓ Coordinated technical committees for data governance and data privacy compliance (GDPR/LGPD), defining clear rules for PII masking and anonymization across analytical environments.

Senior Data Scientist / Senior Data Engineer

2018 – 2021

Large Marketplace & Retail Tech

- ✓ Developed a personalized product recommendation engine responsible for an 18% increase in the platform's Average Order Value (AOV).
- ✓ Architected a Medallion Lakehouse structure integrated with Databricks and PySpark, enabling efficient processing of over 50 TB of daily user behavior logs.
- ✓ Modeled predictive Lifetime Value (LTV) and purchase propensity to drive automated, hyper-targeted marketing campaigns.
- ✓ Built and maintained a central Feature Store, eliminating redundancy in feature engineering workflows between data engineering and data science teams.

Analytics Engineer / Machine Learning Consultant

2015 – 2018

Analytical Consulting & IT Services

- ✓ Provided technical consulting for enterprise clients on modern data architecture definition and cloud migration strategies (AWS/GCP).
- ✓ Constructed robust ETL/ELT pipelines using Apache Airflow and dbt, delivering near real-time analytical reporting.
- ✓ Developed Machine Learning prototypes (PoCs) for churn prediction, customer segmentation, and manufacturing predictive maintenance.
- ✓ Trained client business teams on utilizing self-service BI tools autonomously and interpreting basic statistical propensity models.

CERTIFICATIONS

AWS Certified Data Engineer – Associate / AWS Certified Machine Learning – Specialty

Databricks Certified Professional Data Scientist / Data Engineer

Google Cloud Certified Professional Data Engineer / Professional Machine Learning Engineer

TensorFlow Developer Certificate

NVIDIA Deep Learning Institute Certificate

FinOps Certified Practitioner (FCP)

Professional Scrum Product Owner (PSPO I)

EDUCATION

- ✓ Bachelor of Science in Physics (UNICAMP)
- ✓ Associate Degree in Data Processing (FATEC)
- ✓ MBA in IT Management (UNIP)

LANGUAGES

Advanced English

Native Portuguese

Intermediate Spanish